

Persistent Priors, Preserved Targets: A Stroop-Style Paradigm for Lexical Override

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Abstract

Local definitions can assign a familiar word a temporary meaning while its usual associations remain useful elsewhere. We measure interference from those associations with a matched Stroop-style paradigm. A conflict prompt defines *doctor* as *forest* and compares *forest* with the familiar associate *hospital*. A neutral control replaces *doctor* with a semantically weak word in both the definition and query while keeping *forest* and *hospital* fixed. All 11 model-level means are positive. Aggregate means are also positive for all four conflict families and prompt formats. When no redefinition is present, a stronger preference for the familiar distractor predicts more interference in arbitrary-semantic, polysemy/entity, and domain-definition remappings, while the antonym slope is null. Separately, we patch neutral-control activations into antonym prompts in five 1B–2B models. Patching the defined word, the target word in the definition, and the later query word together restores almost all of the target-minus-distractor margin lost in conflict (normalized recovery $R \in [0.92, 1.06]$). Replacing only that target-word activation with a donor from another item reduces recovery in every tested case. Those donor patches also lower the distractor logit, while the contextual target falls much more than under the same-item patch that restores the margin.

1 Introduction

Language models are often asked to use a familiar word with a temporary local meaning. A legal text can redefine an everyday term, a technical document can assign a common word a domain-specific meaning, and a system prompt can introduce a local convention. In these cases the model should not forget the familiar meaning everywhere. It should use the stipulated meaning where the definition applies while retaining its ordinary lexical knowledge for other contexts. Even when a model produces

the requested answer, it may still assign extra probability to a familiar continuation. This residual interference can be invisible to ordinary accuracy.

We measure that interference with a Stroop-style paradigm (Figure 1). Each item has a familiar word w , a target t supplied by the local definition, and a distractor d favored by the familiar meaning of w . In the running example, the prompt defines *doctor* as *forest*, while *hospital* serves as the familiar distractor. The neutral control replaces *doctor* with a semantically weak word at both the definition and query positions but keeps *forest* and *hospital* unchanged. The target and distractor are therefore identical across conditions. The comparison serves the same control purpose as the classic Stroop task (Stroop, 1935) by holding the required answer pair fixed while changing whether the queried word carries a familiar competing association. The analogy is methodological. We do not assume a perceptual counterpart to lexical interference.

We ask two questions. First, are local redefinitions harder to follow when the model has a stronger ordinary preference for the familiar distractor? Second, once lexical conflict has reduced the target’s advantage over that distractor, which activations from the neutral version of the prompt can restore it? Both questions are measured with the same target-minus-distractor score, $\log P(t) - \log P(d)$.

Across 11 open-weight model settings from the Qwen, Gemma, OLMo, and Mistral families, model-level interference is positive in every setting. The aggregate effect is also positive in all four conflict families and all four prompt formats. Antonym remapping has the smallest mean effect. Arbitrary semantic, polysemy/entity, and domain-definition remappings are 2–4× larger. The regression controls for whether the target already outscores the distractor with no remapping, frequency, tokenization, prompt wrapper, conflict family, model family, scale, and instruction tuning. Even with these

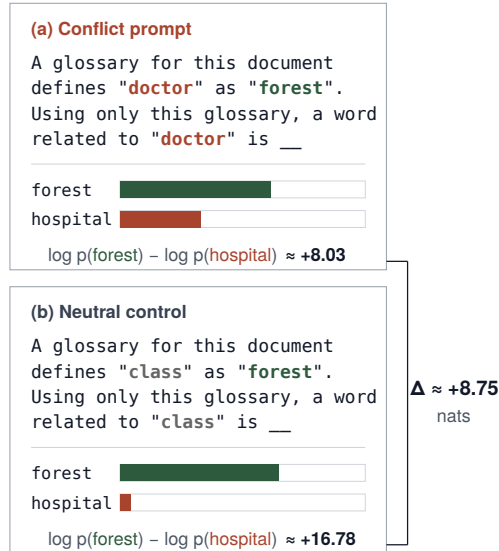


Figure 1: The Stroop-style paradigm. The conflict prompt remaps *doctor* to *forest*, with *hospital* as the familiar distractor. The neutral control replaces *doctor* with *class* at both the definition and query positions while keeping target and distractor fixed. Displayed scores are target-minus-distractor sum log-probability margins under Gemma-2-2B-Instruct. Δ is the neutral minus conflict margin (§3).

controls, the model’s ordinary preference for the distractor predicts more interference. This relation is positive in the three non-antonym families and null within antonym remapping.

We then use residual-stream activation patching in five 1B–2B models to ask what can restore the lost target advantage. The antonym family has a null item-level prior-strength slope, so this experiment addresses a different question from the behavioral regression. Patching the word being defined, the target word in the definition, and the later query word together recovers nearly all of the neutral–conflict gap. At the site and layer chosen for the three-position patch, it outperforms every one- and two-position patch. Its mean advantage over the best pair remains positive on held-out items in all five models. If the activation at the target word in the definition is replaced by one from another item while the defined-word and query patches remain matched, recovery falls for every tested item. The familiar distractor logit still falls under donor patches that fail to recover the margin. The patch using activations from the same item differs mainly by preserving the contextual target logit much better.

Stronger familiar preferences predict more behavioral interference in the three non-antonym fam-

ilies. The antonym patching results show that lowering the familiar distractor logit is not enough to restore the margin when the stipulated target loses too much support.

2 Related Work

Context-parametric conflict and in-context prior competition. A growing literature studies how language models choose between information in the current prompt and knowledge stored in their weights (Longpre et al., 2021; Xie et al., 2024; Kortukov et al., 2024). Xu et al. (2024) survey context–memory, inter-context, and intra-memory conflicts, and recent task-aware evidence shows that conflict behavior depends on what a task requires from context and memory (Sun et al., 2026). Most work operates at the document or claim level through entity reassignment, passage contradiction, or open generation. Related work on in-context learning describes a tension between *task recognition* from priors and *task learning* from demonstrations (Wei et al., 2023; Pan et al., 2023), while irrelevant context can pull predictions away from pretrained expectations (Shi et al., 2023). Under an implicit-Bayesian interpretation (Xie et al., 2022), evidence in the prompt updates a prior over answers to the query. Our paradigm moves this conflict to a single lexical remapping, which allows the target and familiar distractor to be held fixed across the conflict and neutral conditions.

Lexical interference and matched Stroop controls. Psycholinguistic probes have measured lexical interference in pretrained models without the matched Stroop design used here. Ettinger (2020) adapts cloze diagnostics to pretrained models. Kassner and Schütze (2020) show that a misleading prime can pull masked-token predictions toward the prime, and Misra et al. (2020) find that semantic priming in BERT can reverse to interference under stronger sentential constraint. Thota et al. (2026) document failures under locally redefined formal semantics, while Cheng et al. (2026) show that stronger base-model priors predict conflict failure in hypernetwork-based instant adaptation. Repeated in-context lexical substitutions can also shift token semantics across layers (Yona et al., 2026). We induce conflict with an explicit lexical definition and hold the target and distractor fixed under a matched neutral control. This gives an item-level measure of lexical interference during in-context redefinition. A separate prompt with

no redefinition measures the familiar distractor’s advantage and lets us test whether stronger lexical priors predict more interference.

Mechanistic localization, binding, and activation interventions. Activation patching replaces internal activations and measures how the output changes (Vig et al., 2020; Meng et al., 2022; Zhang and Nanda, 2024). It has also been used for causal mediation analysis (Mueller et al., 2026). Circuit analyses localize components that affect model outputs (Wang et al., 2023; Olsson et al., 2022; McDougall et al., 2024). Copy-suppression work shows that internal components can influence a decision by lowering wrong-token logits instead of writing the correct token (McDougall et al., 2024). Within knowledge conflicts, Yu et al. (2023), Ortu et al. (2024), and Jin et al. (2024) identify heads that promote memorized or in-context answers at the entity and document levels. Recent residual-stream patching also identifies causally relevant states under controlled semantic conflicts in code (Abdelsalam et al., 2026), while sparse-autoencoder (SAE) interventions detect and steer context–memory knowledge selection (Zhao et al., 2025). Binding work separately asks how contextual entities remain paired with the correct attributes. Feng and Steinhart (2024) identify binding-ID vectors attached to entities and attributes, and Prakash et al. (2024) find that fine-tuning strengthens entity-tracking mechanisms already present in base models. In a lexical redefinition, the locally defined target must remain associated with a familiar word while that word’s pre-trained associate remains a plausible answer. We patch the word being defined, its stipulated target, and the later query word using activations from the model’s own neutral-control run. This differs from representation-engineering methods that impose model-side directions or fact encodings (Turner et al., 2023; Zou et al., 2023; Hernandez et al., 2024). We then inspect the target and distractor logits separately to see whether the recovered margin comes from preserving the new target, lowering the familiar alternative, or both.

3 The Stroop-style Paradigm

3.1 Stimuli

Each item specifies a familiar word w , a context-defined target t , and a distractor d associated with the familiar meaning of w . The conflict prompt defines w as t and later asks about w .

The neutral prompt defines a semantically weak word n as the same target t and later asks about n . Thus the word appearing in the definition and query changes, while the target and distractor stay fixed (Figure 1). For the running *doctor* item under the glossary wrapper, the conflict prompt reads A glossary for this document defines "doctor" as "forest". Using only this glossary, a word related to "doctor" is, with $t = \textit{forest}$ and $d = \textit{hospital}$. The neutral prompt replaces both occurrences of *doctor* with a neutral word. The relation cue is a synonym of for antonym items and a word related to for the other three families (Appendix A, Table 5). Target and distractor completions have identical surface forms and tokenization across the conflict and neutral conditions. We average over multiple neutral words per item, and the controlled regression includes token-count and frequency covariates (§4.2).

Conflict families. We evaluate four conflict families spanning psycholinguistic and document-style settings. Antonym remapping uses cases such as *small means big* and provides the closest Stroop analogue. Arbitrary semantic remapping includes cases such as *apple means river*. Polysemy/entity remapping includes *jaguar means car*, and domain-definition remapping includes *thread means process*. Table 4 gives examples from each family.

Prompt wrappers. Each item is rendered as a game rule, glossary entry, technical convention, and passage-scoped definition. Table 5 gives the full templates. The wrapper comparison tests whether the effect persists across rule, glossary, technical-document, and passage-scoped framings.

3.2 Behavioral metric

For prompt x , let $S(x) = \log P(t | x) - \log P(d | x)$ denote relative target preference, where $P(\cdot | x)$ is the model’s full-sequence probability under teacher forcing. The Stroop-style interference effect is

$$\Delta = \mathbb{E}_{n \in \mathcal{N}}[S(x_{\text{neutral},n})] - S(x_{\text{conflict}}),$$

where $\mathcal{N}$ is the set of sampled neutral controls for the item. Positive Δ means lexical conflict reduces relative target preference. We report sum log-probability as the primary metric. Under mean per-token scoring, all 11 model means remain positive

(1.32 to 2.65), and the controlled lexical-advantage coefficient remains positive. At the finer cell level, one of the 44 model $\times$ family cells changes sign between the two scoring modes (Mistral-7B-IT on antonym remapping, +0.34 under sum and -0.59 under mean). None of the 44 model $\times$ wrapper cells changes sign.

Worked example. For the running *doctor* item in Figure 1, Gemma-2-2B-Instruct under the glossary wrapper gives $S(x_{\text{conflict}}) \approx +8.03$ with sum log-probability scoring. The displayed neutral control using *class* gives $S(x_{\text{neutral, class}}) \approx +16.78$ and a single-control $\Delta \approx 8.75$. Averaging over six neutral controls gives mean $S(x_{\text{neutral}}) \approx +16.64$ and $\Delta \approx 8.61$.

Lexical-prior strength. We measure the strength of the distractor’s lexical prior with an *ordinary-prior prompt* (e.g., *a word related to doctor is*) that contains no remapping rule. The relation cue matches the family of the conflict prompt, so the prior is measured under the same relation phrase the conflict prompt uses. The lexical-prior advantage of the distractor over the target is $\log P(d) - \log P(t)$ under that prompt, computed per item per model from a single forward pass. This quantity is what we mean by lexical-prior strength throughout.

3.3 Models

The behavioral analysis evaluates 11 open-weight model settings. They include Qwen2.5-1.5B and 7B base/instruct (Yang et al., 2024), Gemma-2-2B and 9B base/instruct (Rivière et al., 2024), OLMo-1B (Groeneveld et al., 2024), and Mistral-7B base/instruct (Jiang et al., 2023). We abbreviate the instruction-tuned variants as “-IT” in figures and tables. Residual-stream activation patching is run on five 1B–2B models. They are Qwen2.5-1.5B base/IT, Gemma-2-2B base/IT, and OLMo-1B (Table 6). These were the models for which TransformerLens (Nanda and Bloom, 2022) loading was reliable and the full residual-stream patching sweep was computationally feasible.

4 Behavioral Results

4.1 Interference across models, conflict families, and prompt wrappers

All 11 model settings have positive aggregate effects (Figure 2), ranging from 1.31 for Gemma-2-9B-IT to 2.71 for Qwen2.5-7B. Positive effects

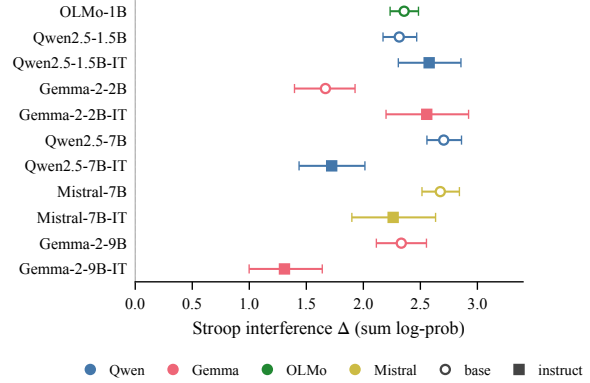


Figure 2: Model-level Stroop interference Δ with bootstrap 95% confidence intervals (sum log-probability scoring). All 11 evaluated model settings have positive aggregate effects, including 7B/9B-class models from four model families.

occur in every model family, in both base and instruction-tuned settings, and at every evaluated model size.

Across conflict families, antonym remapping, the closest psycholinguistic analogue of Stroop conflict, gives the smallest aggregate effect ($\Delta = 1.16$, 95% CI [1.06, 1.26], $n=3520$). The other three families are 2–4 $\times$ larger. Arbitrary semantic remapping gives 3.97 [3.79, 4.15], polysemy/entity 2.66 [2.49, 2.84], and domain-definition 2.54 [2.37, 2.73] (Figure 3(a)). Antonym also contains the only negative model $\times$ family cell and is the only family that uses a synonym-seeking relation cue (Table 9), a design difference considered in Limitations.

Prompt framing changes the magnitude without removing the effect. Mean interference is 3.15 under the game-rule wrapper, 2.06 under glossary, 1.94 under scoped definition, and 1.75 under technical-document prompts (Figure 3(b)). Positive item–prompt observations remain at 78%–81% across wrappers, and 42 of 44 model $\times$ wrapper cells have confidence intervals above zero (Table 10).

4.2 Lexical-prior strength predicts interference

Lexical-prior advantage predicts the size of the interference at the item–prompt level (Figure 4). Pooling all 11 models, bivariate OLS gives a slope of 0.086 ($p < 10^{-9}$), and the decile means increase monotonically. One additional nat of ordinary lexical-prior advantage corresponds to 0.086 nat more interference in the bivariate fit. Across

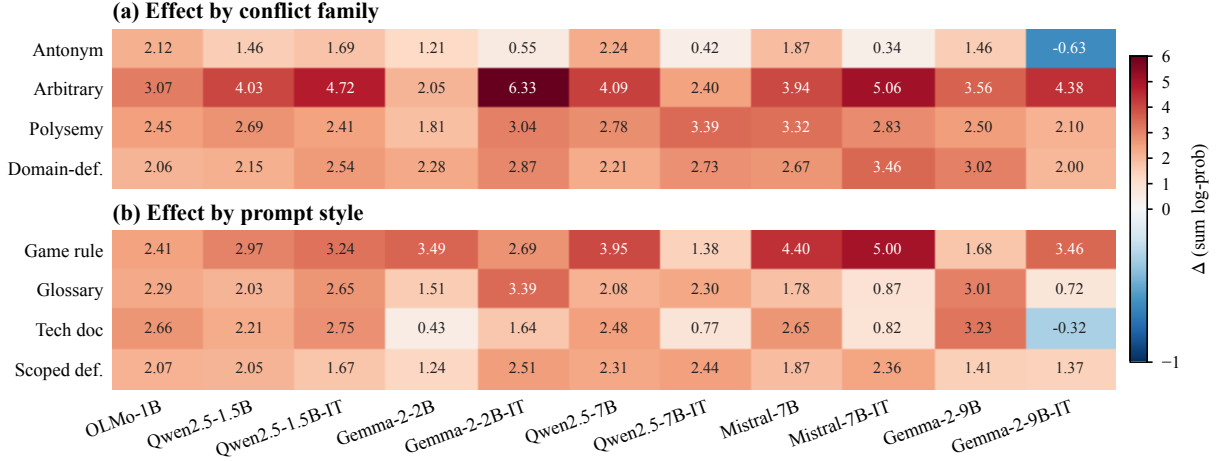


Figure 3: Interference across conflict families and prompt wrappers. **(a)** Effect by conflict family across models. **(b)** Effect by prompt wrapper across models. Aggregate interference is positive in every family and wrapper. Arbitrary-semantic, polysemy/entity, and domain-definition remappings produce larger effects than antonym remapping, which contains the only negative model $\times$ family cell. Cell values are sum log-probability Δ .

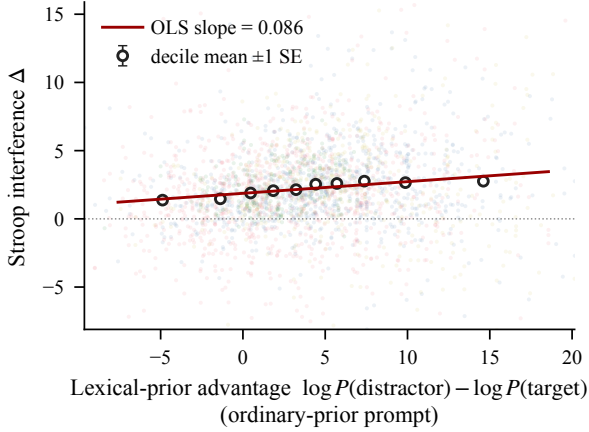


Figure 4: Lexical-prior strength and Stroop interference at the item $\times$ prompt level. Light points show 2,500 subsampled observations across all 11 models. Open circles show decile means with ± 1 SE bars. The red line is the bivariate OLS fit ($n=7,744$). Table 1 reports the controlled regression.

the pooled item–model observations, lexical-prior advantage spans more than 20 nats. The bivariate fit therefore corresponds to about 1.7 nats of interference across that observed range, of the same order as the model-level aggregate effects of 1.31 to 2.71 nats.

The lexical-advantage coefficient remains positive in the controlled regression. The estimate is $+0.114$ ($p < 10^{-9}$), as shown in Table 1. The model includes an answer-prior indicator for whether the target outcores the distractor with no remapping, along with frequency, tokenization, prompt wrapper, conflict family, model family,

Term	Coef.	95% CI	<i>p</i>
Intercept	+0.746	[+0.38, +1.11]	$5.5 \cdot 10^{-5}$
Lexical advantage	+0.114	[+0.08, +0.15]	$4.5 \cdot 10^{-10}$
Answer prior	+0.16	[+0.11, +0.21]	$7.7 \cdot 10^{-11}$
Frequency difference (Zipf)	+0.65	[+0.43, +0.86]	$7.9 \cdot 10^{-9}$
Token-count difference	-0.08	[-0.46, +0.31]	0.69
Instruction-tuned	-0.74	[-1.00, -0.47]	$4.0 \cdot 10^{-8}$
log parameters (B)	-0.32	[-0.44, -0.19]	$4.3 \cdot 10^{-7}$

Table 1: Controlled regression of item $\times$ prompt Stroop interference under sum log-probability scoring ($n=7,744$). Answer prior indicates whether the target outcores the distractor with no remapping. The model also includes fixed effects for conflict family, prompt wrapper, and model family. Standard errors are clustered by item. Table 11 reports three alternative specifications.

scale, and instruction tuning.

Alternative clustering schemes and a mixed-effects specification retain the positive lexical-advantage coefficient. The mixed-effects interval is wider at $[+0.01, +0.21]$ ($p = 0.025$), so the sign is stable while a single common slope is estimated less precisely (Table 11).

The lexical-prior relation varies across conflict families. In the LEXICAL_ADVANTAGE $\times$ CONFLICT_FAMILY specification, the antonym slope is $+0.026$ ($p = 0.235$). Arbitrary semantic, domain-definition, and polysemy/entity remapping give slopes of $+0.202$, $+0.185$, and $+0.150$, all with $p < 10^{-3}$ (Table 12).

5 Mechanistic Analysis

The behavioral regression asks whether a stronger familiar lexical association predicts greater interfer-

ence across items. The activation-patching experiment asks which neutral-control activations can restore the target–distractor margin after lexical conflict reduces it. We run this experiment on antonym remappings in the game-rule wrapper, using single-token items and five 1B–2B models. The antonym family has a null item-level prior-strength slope (§4.2). The patching results therefore show how the target–distractor margin can be restored for these antonym items. They do not test why prior strength predicts interference in the other three families.

5.1 Patching the defined word, its target, and the query

For each item we compare a conflict prompt with a neutral-control prompt that has the same target, distractor, and token length. The conflict prompt uses the familiar word in both the definition and the later query. The neutral prompt uses a semantically weak word in those two positions instead. Eligible items have single-token familiar words, targets, and distractors and positive interference for that model. This yields 32 items per model and 14 for Gemma-2-2B-IT (Appendix E).

We copy residual-stream activations from the neutral prompt into the corresponding conflict prompt. We patch three positions. The first is the word being defined (the definition subject), the second is the target word supplied by the definition (the definition target), and the third is the later query word. We patch each position alone, all three pairs, and all three positions together. The input tokens at the definition subject and query positions differ between the two prompts. The definition-target token is identical. Patching the target position therefore transfers the activation that this same token acquired in the neutral context. Its token identity does not change.

Recovery is measured at the final prompt position using the target-minus-distractor logit margin. Let $M = \ell_t - \ell_d$ denote this margin. Following Zhang and Nanda (2024), $R = (M_{\text{patched}} - M_{\text{conflict}}) / (M_{\text{neutral}} - M_{\text{conflict}})$. $R = 1$ reproduces the neutral-control margin and $R = 0$ leaves the conflict margin unchanged. Values above 1 overshoot the neutral-control margin. Values below 0 move the margin farther below the conflict baseline.

The subject and query patches act at the two positions where the prompt tokens themselves differ. Patching those positions early in the network can therefore approach replacement of the differ-

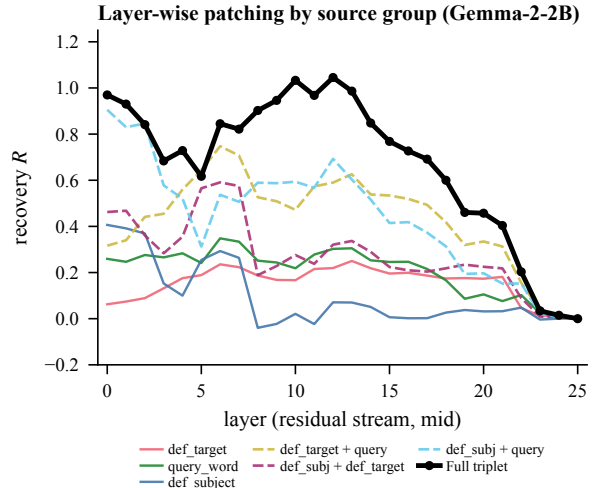


Figure 5: Layer-wise recovery for each patched position combination on Gemma-2-2B at the residual-stream site selected by the three-position patch (mid-block). The three-position patch exceeds every one- and two-position patch across the mid-network layers shown. Table 2 reports all five models used for activation patching at their selected sites and layers.

Model	Site	L	R_{tp}	R_{pair}	ΔR
Qwen2.5-1.5B	input	3	0.99	0.89	+0.11
Qwen2.5-1.5B-IT	input	15	0.92	0.59	+0.33
Gemma-2-2B	mid-block	12	1.05	0.69	+0.35
Gemma-2-2B-IT	input	3	1.04	0.68	+0.36
OLMo-1B	input	2	1.06	0.87	+0.19

Table 2: Recovery from the three-position patch and the best two-position patch at the site and layer selected by the three-position patch. Input denotes block input and mid-block denotes post-attention. R_{tp} is recovery from the three-position patch, and R_{pair} is recovery from the best two-position patch at the same comparison point. The best pair patches the defined word and the later query word in all five models.

ing input tokens. The definition-target patch is different because that token is the same in both prompts. To test whether its item-matched activation contributes beyond the subject and query patches, we keep those two patches matched to the current item and replace only the definition-target activation with one from another item. A second control replaces all three patched activations with activations from a different item. Implementation details are in Appendix E.

Patching all three positions recovers nearly all of the neutral–conflict gap. At the residual-stream site and layer selected for the three-position patch, mean recovery ranges from 0.92 to 1.06 across the five models, with an overall mean of 1.01. At that same site and layer, the three-position patch out-

performs every one- and two-position patch. Its advantage over the best pair ranges from +0.11 to +0.36. The strongest pair is always definition subject plus query word, so adding the definition-target activation improves recovery without changing the comparison site (Figure 5 and Table 2). Results for every position combination are reported in Table 13.

The three-position advantage also appears on held-out items. Across 200 discovery/held-out splits per model, the mean held-out advantage over the best pair ranges from +0.11 to +0.39, and 83%–100% of splits have a positive difference. The split-level intervals are wider in the two instruction-tuned models (Table 15). If each position combination is instead allowed to choose its own best site and layer, the definition-subject plus query pair can recover strongly near the input. Because these are exactly the two token positions that differ between the neutral and conflict prompts, early patching there approaches replacement of the input difference. Appendix F reports that alternative comparison.

Patches taken from a different item do not restore the margin. Across 50 random item permutations per model, mean recovery ranges from -0.58 to -4.16 , and even the best permutation remains below zero in every model (Table 16).

5.2 Recovery falls when the definition-target activation comes from another item

The definition-subject plus query-word patch is the strongest two-position patch in every model. We therefore keep those two neutral activations matched to the current item and change only the activation patched at the definition target. When that activation comes from another item, recovery decreases in all 142 tested item–model cases, and the mean recovery after the swap is negative in all five models. For every item, the matched three-position patch beats the donor swap by at least 1.9 nats on the final target-minus-distractor margin. At the model level, the matched-minus-swap difference ranges from +1.54 to +5.26 recovery units (Table 14). The neutral-control activations at the defined-word and query positions are therefore insufficient by themselves. Replacing the definition-target activation with a donor sharply reduces recovery even though those two positions remain matched to the current item.

5.3 The same-item patch preserves the contextual target better than donor controls

Why does the donor swap fail? The target-minus-distractor margin can improve if the contextual target logit rises, the familiar distractor logit falls, or both. Table 3 separates these two logits for the matched three-position patch, the definition-target donor swap, and the all-position item-mismatched control.

All three interventions lower the distractor logit. Under the same-item three-position patch, it falls by 3.1 to 11.1 nats. Under the donor swap and the all-position mismatch, it falls by 5.3 to 13.7 nats. A large distractor-logit decrease therefore occurs both in the same-item patch that restores the margin and in donor patches that do not.

The target logit shows a different pattern. In four models, the same-item patch changes it by between +0.17 and -1.78 nats. In Qwen2.5-1.5B-IT it falls by 6.97 nats. Under the donor swap and all-position mismatch, the target falls by 7.6 to 16.9 nats. Within each model, replacing the same-item definition-target activation causes an additional target-logit decrease of 6.3 to 15.1 nats. The distractor logit changes by only an additional 2.1 to 2.7 nats. The two mismatched interventions differ by at most 0.6 nat in total margin change. The same-item three-position patch preserves the contextual target much better. The distractor-logit decrease is at least as large in the mismatched controls.

6 Discussion

An explicit local definition does not remove the influence of a familiar lexical association. In arbitrary-semantic, polysemy/entity, and domain-definition items, interference is larger when the model’s ordinary-prior prompt gives the familiar distractor a larger advantage over the new target. The activation-patching experiment uses antonym items, where this item-level slope is null. It addresses recovery after conflict and does not explain why stronger priors predict more interference in the other three families.

The familiar distractor logit falls under the same-item three-position patch, the definition-target donor swap, and the patch that takes all three activations from another item. The target logit falls much less under the same-item patch than under either donor control. Lowering the familiar distractor can

Model	Matched (M)			Swap (S)			Mismatch (R)		
	$\Delta\ell_t$	$\Delta\ell_d$	Δm	$\Delta\ell_t$	$\Delta\ell_d$	Δm	$\Delta\ell_t$	$\Delta\ell_d$	Δm
Qwen2.5-1.5B	-1.39	-3.13	+1.74	-12.79	-5.34	-7.45	-12.68	-5.41	-7.27
Qwen2.5-1.5B-IT	-6.97	-11.05	+4.08	-16.48	-13.70	-2.77	-16.26	-13.34	-2.91
Gemma-2-2B	+0.17	-3.72	+3.89	-8.04	-6.02	-2.03	-7.95	-5.81	-2.14
Gemma-2-2B-IT	-1.78	-5.60	+3.83	-16.92	-8.03	-8.89	-15.69	-7.40	-8.30
OLMo-1B	-1.38	-3.41	+2.03	-7.68	-5.50	-2.19	-7.56	-5.33	-2.23

Table 3: Logit decomposition for the same-item three-position patch (M), definition-target donor swap (S), and the control that patches all three positions from a different item (R). $\Delta m = \Delta\ell_t - \Delta\ell_d$. The distractor logit decreases in all three conditions. The target logit decreases much more when the matched definition-target activation is replaced. The two mismatched controls differ by at most 0.6 nat in margin change.

therefore fail to restore the margin when the locally defined alternative loses too much support by the final prediction.

A local definition does not require the model to erase its ordinary lexical knowledge. If *doctor* is defined as *forest* for one passage, the usual relation between *doctor* and *hospital* should remain available in other contexts. The narrower requirement is that this familiar relation should not pull the current answer away from *forest* where the definition applies. The model must retain ordinary lexical knowledge without letting it determine the answer in the local scope.

In the resource-rational account of [Zhu and Griffiths \(2025\)](#), limited computation tempers the likelihood while leaving the prior exponent at one. Such a rule can sensibly preserve prior influence when new evidence is uncertain. A local definition has a different status because it specifies how the term should be interpreted within a stated scope. A system that blends this stipulation with its prior as though the stipulation were uncertain evidence could leave residual competition of the kind measured here. Our experiments do not estimate updating exponents or show that language models implement this algorithm. The comparison shows why a policy suited to uncertain evidence can be ill-suited when the prompt explicitly stipulates which mapping applies. Related behavioral work on conservative updating documents a similar persistence of prior influence ([Edwards, 1968](#); [Grether, 1992](#)).

An intervention can lower the familiar distractor while reducing the contextual target even more. Reporting the distractor-logit decrease alone would miss interventions that lower the familiar answer but fail to restore the target margin. Target and distractor logits should therefore be examined separately when evaluating interventions on local definitions.

7 Conclusion

Familiar lexical associations continue to bias target-versus-distractor probabilities after an explicit local redefinition. Across 11 open-weight model settings, model-level interference is positive in every setting, and the aggregate effect is positive in all four conflict families and all four prompt wrappers. In the three non-antonym families, stronger ordinary preference for the familiar distractor predicts greater interference. The ordinary-prior score therefore provides an item-level predictor of lexical interference in these families.

On antonym items in five 1B–2B models, we patch neutral-control activations into the conflict prompt. Patching the word being defined, the target word in the definition, and the later query word together recovers nearly all of the neutral–conflict gap. Replacing only the activation at the target word in the definition with one from another item reduces recovery in every tested case. The familiar distractor logit falls both when the patched activations come from the current item and when they come from donor items. The patch using activations from the same item preserves the contextual target logit much better. That patch restores the margin, whereas the donor interventions do not, even though all of them lower the distractor logit.

Limitations

The paradigm uses short controlled prompts, English lexical stimuli, and target-versus-distractor likelihood scoring. The glossary, technical-document, and scoped-definition wrappers approximate local definitions without reproducing the length and complexity of realistic documents, and the experiments do not show how the same interference appears in unconstrained generation.

The activation-patching experiments cover five 1B–2B models and use antonym remapping in the game-rule wrapper, with single-token items

that show positive interference. The behavioral prior-strength predictor is positive in the three non-antonym families and null within antonyms, so the current patching results characterize recovery for those antonym items. Held-out splits reduce dependence on the site and layer selected by the three-position patch, although uncertainty is wider in the instruction-tuned models. The definition-target swap shows that restoring the two word positions that differ between the neutral and conflict prompts is insufficient when the definition-target activation is replaced by a donor. Because donor items generally also carry different target tokens, a same-target donor design would be needed to isolate contextual binding from target-token identity within that activation.

Antonym remapping also uses a different relation cue. Its a synonym of wording preserves the intended conflict, whereas the other three families use a broader relation cue. Varying the cue while holding the remapping relation fixed would separate cue effects from family effects.

Ethical Considerations

This work uses synthetic stimuli and open-weight models and involves no human subjects or sensitive data. The paradigm is intended as a diagnostic for systems that must respect context-specific definitions in settings such as law, medicine, and technical documentation.

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A Stimulus Construction and Prompt Templates

Behavioral set. The behavioral stimulus set contains 176 unique lexical items, each rendered under four prompt styles, yielding 704 item–prompt observations per model setting. The set contains 80 antonym items, 36 arbitrary semantic remappings, 30 polysemy/entity remappings, and 30 domain-definition remappings. Targets and distractors are chosen to be single tokens under each model’s tokenizer whenever possible. When they are not, the target/distractor token-count difference is included as a regression covariate (Table 1). The full stimulus list is released in `behavior_external_validity_stimuli.csv`.

Neutral controls. Neutral controls replace the familiar word at both its definition and query occurrences with semantically weak words drawn from a per-item candidate pool of 6–8 words, screened to be semantically weak in context (e.g., *thing*, *object*, *item*, *label*, *symbol* for some items, and *class*, *field*, *point*, *record*, *slot*, *type* for items where those generic placeholders are themselves contextually plausible). Behavioral experiments average $K=6$

controls per item. Mechanistic experiments select a single representative clean prompt closest to the item’s neutral mean from $K=8$ candidates, giving a deterministic clean run for activation patching. The full per-item neutral lists are released in the stimulus CSV.

Query	Target	Distractor
<i>Antonym remapping (n=80)</i>		
<i>small</i>	<i>big</i>	<i>tiny</i>
<i>large</i>	<i>small</i>	<i>big</i>
<i>hot</i>	<i>cold</i>	<i>warm</i>
<i>cold</i>	<i>hot</i>	<i>chilly</i>
<i>happy</i>	<i>sad</i>	<i>glad</i>
<i>sad</i>	<i>happy</i>	<i>unhappy</i>
<i>Arbitrary semantic remapping (n=36)</i>		
<i>apple</i>	<i>river</i>	<i>fruit</i>
<i>banana</i>	<i>mountain</i>	<i>fruit</i>
<i>tiger</i>	<i>pencil</i>	<i>animal</i>
<i>doctor</i>	<i>forest</i>	<i>hospital</i>
<i>teacher</i>	<i>ocean</i>	<i>school</i>
<i>piano</i>	<i>desert</i>	<i>music</i>
<i>Polysemy / entity remapping (n=30)</i>		
<i>apple</i>	<i>music</i>	<i>iPhone</i>
<i>Python</i>	<i>programming</i>	<i>snake</i>
<i>Java</i>	<i>programming</i>	<i>coffee</i>
<i>jaguar</i>	<i>car</i>	<i>animal</i>
<i>mercury</i>	<i>metal</i>	<i>planet</i>
<i>bass</i>	<i>music</i>	<i>fish</i>
<i>Domain-definition remapping (n=30)</i>		
<i>port</i>	<i>socket</i>	<i>harbor</i>
<i>thread</i>	<i>process</i>	<i>sewing</i>
<i>cell</i>	<i>spreadsheet</i>	<i>biology</i>
<i>token</i>	<i>symbol</i>	<i>coin</i>
<i>driver</i>	<i>software</i>	<i>person</i>
<i>kernel</i>	<i>system</i>	<i>seed</i>

Table 4: Six examples per conflict family. The conflict definition takes the form `<Query>` means `<Target>` in the `game_rule` wrapper, with the other three wrappers paraphrasing it (Table 5). The full stimulus set (176 items) is released in `behavior_external_validity_stimuli.csv`.

Prompt templates. The four prompt wrappers are listed in Table 5. Each wrapper embeds the conflict definition and the query in a different framing. Behavioral experiments instantiate all four wrappers. Mechanistic experiments use the `game_rule` wrapper from the same template family.

game_rule
Complete the answer with one word or phrase only. In this game, {subject} means {meaning}. Question: In this game, {relation_phrase} {query} is Answer:
glossary
A glossary for this document defines "{subject}" as "{meaning}". Using only this glossary, {relation_phrase} "{query}" is
technical_document
In the following technical document, the term "{subject}" refers to "{meaning}". According to the document, {relation_phrase} "{query}" is
definition_scope
For this passage only, interpret "{subject}" as "{meaning}". Under this definition, {relation_phrase} "{query}" is

Table 5: Prompt wrappers. {subject} and {meaning} are filled in from the conflict definition. {query} is the query word. {relation_phrase} is “a synonym of” for the antonym family and “a word related to” for the other three conflict families. The wrappers correspond to the prose names *game-rule*, *glossary*, *technical-document*, and *scoped-definition*.

B Model Set and Tokenization

Table 6 lists all model settings, with a flag indicating whether each is used for the behavioral analysis, the mechanistic analysis, or both. The 7B/9B-class models are included in the broader behavioral sweep. Activation patching is run on Qwen2.5-1.5B base/IT, Gemma-2-2B base/IT, and OLMo-1B, where residual-stream interventions are computationally feasible and TransformerLens (Nanda and Bloom, 2022) loading is reliable.

Model	Family	Beh.	Mech.
Qwen2.5-1.5B	Qwen	✓	✓
Qwen2.5-1.5B-IT	Qwen	✓	✓
Qwen2.5-7B	Qwen	✓	
Qwen2.5-7B-IT	Qwen	✓	
Gemma-2-2B	Gemma	✓	✓
Gemma-2-2B-IT	Gemma	✓	✓
Gemma-2-9B	Gemma	✓	
Gemma-2-9B-IT	Gemma	✓	
OLMo-1B	OLMo	✓	✓
Mistral-7B	Mistral	✓	
Mistral-7B-IT	Mistral	✓	

Table 6: Open-weight model settings. “IT” is the instruction-tuned variant. “Beh.” marks inclusion in the behavioral sweep ($n=11$). “Mech.” marks inclusion in the activation-patching analysis ($n=5$).

Tokenization. Target and distractor tokenization is identical across conflict and neutral conditions. The word used at the definition-subject and query positions differs by design. The controlled regression includes target/distractor token-count and frequency covariates (Table 1).

C Full Behavioral Results Tables

Table 7 gives aggregate interference by conflict family and prompt wrapper. The released per-cell summaries are `behavior_summary_by_model_conflict_type.csv` and `behavior_summary_by_model_prompt_style.csv`.

	n	Δ	95% CI	Pos.
<i>By conflict family</i>				
Antonym	3520	1.16	[1.06, 1.26]	0.70
Arbitrary	1584	3.97	[3.79, 4.15]	0.92
Polysemy/ent.	1320	2.66	[2.49, 2.84]	0.86
Domain-def.	1320	2.54	[2.37, 2.73]	0.84
<i>By prompt wrapper</i>				
Game-rule	1936	3.15	[2.92, 3.38]	0.78
Glossary	1936	2.06	[1.92, 2.19]	0.81
Technical-document	1936	1.75	[1.64, 1.88]	0.80
Scoped-definition	1936	1.94	[1.82, 2.06]	0.80

Table 7: Aggregate Stroop interference Δ by conflict family and by prompt wrapper (sum log-probability). “Pos.” is the fraction of item–prompt–model observations with positive Δ .

Table 8 gives the model-level aggregate effects underlying Figure 2.

Model	Size	Δ	95% CI	Pos.
OLMo-1B	1.0	2.36	[2.24, 2.48]	0.95
Qwen2.5-1.5B	1.5	2.31	[2.17, 2.47]	0.92
Qwen2.5-1.5B-IT	1.5	2.58	[2.31, 2.86]	0.80
Gemma-2-2B	2.0	1.67	[1.40, 1.93]	0.71
Gemma-2-2B-IT	2.0	2.56	[2.20, 2.92]	0.75
Qwen2.5-7B	7.0	2.71	[2.56, 2.86]	0.93
Qwen2.5-7B-IT	7.0	1.72	[1.44, 2.01]	0.72
Mistral-7B	7.0	2.67	[2.51, 2.84]	0.91
Mistral-7B-IT	7.0	2.26	[1.90, 2.63]	0.66
Gemma-2-9B	9.0	2.33	[2.11, 2.55]	0.81
Gemma-2-9B-IT	9.0	1.31	[1.00, 1.64]	0.61

Table 8: Model-level Stroop interference Δ (sum log-probability), $n=704$ item–prompt observations per model. Mean per-token scoring gives the same conclusion that all 11 settings are positive at the model level (released in `behavior_summary_by_model.csv`).

Tables 9 and 10 give the numerical values behind Figure 3 for every model×family and model×wrapper cell.

Model	Ant.	Arb.	Pol.	Dom.
OLMo-1B	2.12	3.07	2.45	2.06
Qwen2.5-1.5B	1.46	4.03	2.69	2.15
Qwen2.5-1.5B-IT	1.69	4.72	2.41	2.54
Gemma-2-2B	1.21	2.05	1.81	2.28
Gemma-2-2B-IT	0.55	6.33	3.04	2.87
Qwen2.5-7B	2.24	4.09	2.78	2.21
Qwen2.5-7B-IT	0.42	2.40	3.39	2.73
Mistral-7B	1.87	3.94	3.32	2.67
Mistral-7B-IT	0.34	5.06	2.83	3.46
Gemma-2-9B	1.46	3.56	2.50	3.02
Gemma-2-9B-IT	-0.63	4.38	2.10	2.00

Table 9: Per-cell Δ by model and conflict family (sum log-probability), corresponding to Fig. 3(a). Ant. denotes antonym, Arb. arbitrary semantic, Pol. polysemy/entity, and Dom. domain-definition.

Model	Game	Gloss.	Tech.	Scope
OLMo-1B	2.41	2.29	2.66	2.07
Qwen2.5-1.5B	2.97	2.03	2.21	2.05
Qwen2.5-1.5B-IT	3.24	2.65	2.75	1.67
Gemma-2-2B	3.49	1.51	0.43	1.24
Gemma-2-2B-IT	2.69	3.39	1.64	2.51
Qwen2.5-7B	3.95	2.08	2.48	2.31
Qwen2.5-7B-IT	1.38	2.30	0.77	2.44
Mistral-7B	4.40	1.78	2.65	1.87
Mistral-7B-IT	5.00	0.87	0.82	2.36
Gemma-2-9B	1.68	3.01	3.23	1.41
Gemma-2-9B-IT	3.46	0.72	-0.32	1.37

Table 10: Per-cell Δ by model and prompt wrapper (sum log-probability), corresponding to Fig. 3(b). Game denotes game-rule, Gloss. glossary, Tech. technical-document, and Scope scoped-definition.

D Regression Sensitivity Analyses

Full fixed-effect coefficients and interaction estimates are released in `behavior_control_regressions.csv`. Table 11 reports the lexical-advantage coefficient under item clustering, model clustering, two-way clustering, and a mixed-effects specification with item- and model-level random effects. The corresponding output is in `behavior_regression_robustness.csv`. Mean log-probability scoring yields the same qualitative pattern.

Specification	Coef.	95% CI	p
Item-clustered	+0.114	[+0.08, +0.15]	$4.5 \cdot 10^{-10}$
Model-clustered	+0.114	[+0.07, +0.16]	$3.7 \cdot 10^{-6}$
Two-way clustered	+0.114	[+0.06, +0.17]	$4.8 \cdot 10^{-5}$
Mixed-effects	+0.114	[+0.01, +0.21]	0.025

Table 11: Sensitivity of the LEXICAL_ADVANTAGE coefficient under sum log-probability scoring ($n=7,744$) to alternative clustering and mixed-effects specifications.

Family	Coef.	95% CI	p
Antonym (baseline)	+0.026	[-0.017, +0.070]	0.235
Arbitrary semantic	+0.202	[+0.13, +0.28]	$1.5 \cdot 10^{-7}$
Domain-definition	+0.185	[+0.09, +0.28]	$1.5 \cdot 10^{-4}$
Polysemy/entity	+0.150	[+0.08, +0.22]	$5.8 \cdot 10^{-5}$

Table 12: Family-specific lexical-prior slopes from the LEXICAL_ADVANTAGE $\times$ CONFLICT_FAMILY interaction specification (released as `behavior_control_regressions.csv`, `regression_type_interactions`). The antonym baseline slope is null. The other three families show positive lexical-prior effects.

E Mechanistic Implementation

Prompt construction and item selection. For activation patching, we call the neutral-control prompt the clean run and the lexical-conflict prompt the corrupted run. They share the same target and distractor (§5.1). The clean prompt is selected from $K=8$ neutral candidates as the one whose target-versus-distractor score is closest to the item’s neutral mean, giving a deterministic clean run for activation patching.

An antonym item is included in the activation-patching analysis for a given model when its query word, target, and distractor are single tokens under that model’s tokenizer, its clean and corrupted prompts have equal token length, and its item-level interference is positive. The positive-interference criterion ensures that the neutral-control margin exceeds the conflict margin, so the normalized recovery score has a positive gap to restore. Up to 32 eligible items are used per model, giving 32 items for Qwen2.5-1.5B base and IT, Gemma-2-2B, and OLMo-1B and 14 for Gemma-2-2B-IT, which has the weakest antonym effect among the five models used for activation patching (Table 9).

Patching sites, positions, and recovery. We use three residual-stream sites within each transformer block. Block input is recorded as `hook_resid_pre`. Mid-block, after attention and before the MLP, is recorded as `hook_resid_mid`. Block output is recorded as `hook_resid_post`. The main paper uses the readable names, while the released CSVs retain the TransformerLens field names.

The three patched positions are stored as `def_subject`, `def_target`, and `query_word`. The CSV field `source_group` records which combination of these positions is patched, and the three-position intervention is named `source_all`. The *final position* is

the final prompt position at which the target-minus-distractor logit difference is read. The definition-target swap tests what changes when `def_target` comes from another item while the other two patched positions remain matched to the current item (§5).

The patching score is the final-position target-minus-distractor logit difference $M = \ell_t - \ell_d$. Recovery is $R = (M_{\text{patched}} - M_{\text{corrupt}}) / (M_{\text{clean}} - M_{\text{corrupt}})$, the normalized metric of Zhang and Nanda (2024). A value of $R=1$ reproduces the clean-run margin, while $R=0$ leaves the corrupted-run margin unchanged. Values above 1 overshoot the clean margin, and values below 0 move the margin below the corrupted baseline.

Held-out validation and different-item controls. For each model we run 200 repeated splits. Each split chooses a discovery item set, selects the best site, layer, and patched-position combination on the discovery items, and evaluates that choice on held-out items. We record $\Delta R = R_{\text{triplet}} - R_{\text{best pair}}$. Results are summarized in Table 15.

For each model we also run 50 control perturbations using neutral-control activations from a randomly chosen *different* item while keeping the current conflict item’s target and distractor fixed. These patches do not restore the target–distractor margin and often reduce it further (Table 16).

F Mechanistic Controls and Validation

Position combinations at the site and layer selected by the three-position patch. Table 13 reports every patched-position combination at the site and layer selected by the three-position patch. In the table, `def_s` is the defined-word position, `def_t` is the definition-target position, and `query` is the later query position. The released residual-patching CSV contains every position combination, layer, and residual-stream site.

Group (site, L)	Q1.5 input, 3	Q1.5-IT input, 15	G2 mid-block, 12	G2-IT input, 3	OLMo input, 2
<code>def_s</code>	−2.46	−0.09	+0.07	−3.33	−0.02
<code>def_t</code>	+0.03	−1.99	+0.22	+0.34	+0.06
<code>query</code>	−2.35	+0.35	+0.30	−1.20	−0.58
<code>def_s + def_t</code>	−2.31	−1.92	+0.32	−3.75	+0.09
<code>def_s + query</code>	+0.89	+0.59	+0.69	+0.68	+0.87
<code>def_t + query</code>	−2.20	−0.09	+0.59	−1.02	−0.50
Triplet	0.99	0.92	1.05	1.04	1.06

Table 13: Recovery for each patched-position combination at the site and layer selected by the three-position patch. The three-position patch exceeds every one- and two-position patch in all five models. Q1.5 = Qwen2.5-1.5B, G2 = Gemma-2-2B, and the IT suffix denotes instruction tuning.

Allowing each position combination to choose its own layer. When each patched-position combination selects its own site and layer, the definition-subject plus query-word pair can reach high recovery near the input. These are the two token positions that differ between the clean and corrupted prompts, so early patching approaches replacement of the input difference. The main text therefore compares all position combinations at the site and layer selected by the three-position patch and uses held-out split validation as a separate check. All per-combination and per-layer values are in `mechanism_source_residual_patches.csv`.

Definition-target swap control. Donor definition-target activations are drawn from the same pool of eligible items used for the same-item three-position patch. We run 10 donor swaps per item at each model’s selected site and layer. Table 14 reports recovery from the same-item patch, mean recovery after the donor swap, and the fraction of items whose mean swap recovery falls below the same-item value (*every* item in every model). The swap and item-mismatched controls produce similar margin changes (§5, Table 3).

Model	(site, L)	n	Matched	Swap	Frac. below
Qwen2.5-1.5B	input, 3	32	+0.99	−4.26	1.00
Qwen2.5-1.5B-IT	input, 15	32	+0.92	−0.62	1.00
Gemma-2-2B	mid-block, 12	32	+1.05	−0.54	1.00
Gemma-2-2B-IT	input, 3	14	+1.04	−2.43	1.00
OLMo-1B	input, 2	32	+1.06	−1.14	1.00

Table 14: Definition-target swap control. Definition subject and query activations remain matched to the item while the definition-target activation is replaced by a donor item’s activation. Swap is the mean over 10 donors per item. Frac. below is the fraction of items whose mean swap recovery falls below recovery from the same-item three-position patch.

Split-validation summary. Table 15 reports the 200-fold split-validation results for each activation-patching model. It gives the held-out mean $\Delta R = R_{\text{triplet}} - R_{\text{pair}}$, its 95% split-level confidence interval, and the fraction of splits with positive ΔR (cf. §5).

Model	Splits	Mean ΔR	95% CI	Pos.
Qwen2.5-1.5B	200	+0.11	[+0.00, +0.34]	0.975
Qwen2.5-1.5B-IT	200	+0.31	[−0.01, +0.56]	0.950
Gemma-2-2B	200	+0.39	[+0.03, +0.66]	0.975
Gemma-2-2B-IT	200	+0.19	[−0.11, +0.58]	0.825
OLMo-1B	200	+0.18	[+0.08, +0.28]	1.000

Table 15: Held-out split-validation summary for each activation-patching model (`mechanism_split_validation.csv`, summary in `mechanism_split_validation_summary.csv`). **Pos.** is the fraction of splits with positive ΔR .

Different-item control summary. Table 16 summarizes the 50 controls that patch all three positions with activations from a different item. Recovery from the same-item three-position patch is near or above 1 for every model, while mean recovery under the different-item controls is negative and the highest single control value (**Ctrl. max**) remains below zero in every model.

Model	Matched	Ctrl. mean	Ctrl. min	Ctrl. max
Qwen2.5-1.5B	0.99	−4.16	−4.90	−3.35
Qwen2.5-1.5B-IT	0.92	−0.66	−0.88	−0.40
Gemma-2-2B	1.05	−0.58	−0.80	−0.30
Gemma-2-2B-IT	1.04	−2.27	−3.10	−1.33
OLMo-1B	1.06	−1.16	−1.38	−0.87

Table 16: Different-item control summary for each activation-patching model (`mechanism_random_controls.csv`, 50 permutations per model). Recovery from the same-item three-position patch exceeds the highest control value in every model.

G Reproducibility and Released Artifacts

Repository and released artifacts. The implementation, end-to-end reproduction notebook, released CSV outputs, figures, Markdown summaries, and package documentation are available at <https://github.com/henryhyw/priors-persist-through-suppression>. The repository README maps each released CSV to the figure or table it supports. The code is MIT-licensed. Model checkpoints, TransformerLens, and Hugging Face tooling remain subject to their upstream licenses and access requirements.

Compute, tooling, and tokenizer handling. Reproduction uses a single A100-class GPU and requires forward evaluation, activation caching, and patching without model training or fine-tuning. Behavioral evaluation uses `bfloat16` where available, while mechanistic experiments use `float32` for activation caching and residual-stream arithmetic. Mechanistic interventions use TransformerLens (Nanda and Bloom, 2022). Gated checkpoints require the corresponding Hugging Face access token. For tokenizers without a valid beginning-of-sequence token, including the Qwen family, we disable automatic BOS insertion and score with `add_special_tokens=False`.

Determinism and data. Behavioral evaluation is deterministic given fixed model weights and tokenizer handling. Mechanistic split validation and item-mismatched controls use the fixed random seed recorded in the notebook, and the released CSVs are the outputs of that seeded run. Released data consist of synthetic prompts, lexical-item lists, derived numerical outputs, figures, and summaries. No human-subject or private-user data were collected.